

Project Design Phase

Problem – Solution Fit Template

Date	18 June 2026
Team ID	
Project Name	ShopEZ – MERN Stack E-Commerce Platform
Maximum Marks	2 Marks

Problem – Solution Fit Template:

Define CS, fit into CL	1. CUSTOMER SEGMENT(S) CS - Home cooks & food enthusiasts - Busy professionals & students - Health-conscious individuals - People with dietary restrictions - Users seeking personalized meal ideas - Food bloggers & content creators	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES CL - Limited time to plan and cook - Lack of ingredient knowledge - Difficulty finding suitable recipes - Budget constraints for premium ingredients - Overwhelmed by too many options - No personalized cooking guidance	5. AVAILABLE SOLUTIONS PROS & CONS AS Existing Solutions: - Allrecipes - Yummly - Tasty - Spoonacular - ChatGPT / Gemini (manual prompts) - YouTube Cooking Channels Limitations: - Generic recommendations - Limited personalization - Lack of real-time guidance - No voice & visual interaction - Scattered user experience - Subscription barriers
	Focus on PR, tap into BE, understand RC	2. PROBLEMS / PAINS + ITS FREQUENCY PR - Don't know what to cook with available ingredients - Recipes not personalized to health or taste - Hard to follow recipes while cooking - Lack of real-time cooking support - Multiple apps for recipes, notes, and planning - Difficulty in tracking nutrition & calories - No voice or visual cooking assistance - Time-consuming meal planning	9. PROBLEM ROOT / CAUSE RC - Recipes are not context-aware - No integration of user preferences & history - Complex user interfaces - Lack of AI-powered personalization - No real-time, step-by-step guidance - Siloed data and poor state management - Lack of multimodal interaction (voice, text, visual) - Limited cloud integration for scalability
Identify strong TR & EM		3. TRIGGERS TO ACT TR - Hunger / Meal planning - Available ingredients at home - Diet or health goals - Special occasions & events - Trending recipes & food ideas - Time-saving cooking needs	10. YOUR SOLUTION SL RecipeCraft AI is an AI-powered recipe generation system that delivers personalized, real-time cooking guidance through multimodal interactions (voice, text, visual). Key Features: - AI-powered recipe generation (personalized & context-aware) - Real-time step-by-step cooking guidance - Voice, text & visual interactions - Ingredient-based & goal-based recommendations - Nutrition analysis & calories tracking (visualized) - Save, organize & share recipes - Smart meal planner & grocery suggestions - Secure authentication & user personalization RecipeCraft AI transforms the way you discover, cook, and enjoy food with intelligent assistance at every step.
	4. EMOTIONS BEFORE / AFTER EM BEFORE - Unsure what to cook - Overwhelmed - Frustrated - Running out of ideas - Worried about nutrition AFTER - Confident - Inspired - Satisfied - Healthy & happy - Excited to cook more	TECH STACK (MERN + CLOUD)	